Utkarsh Tyagi

Audio & Multimodal AI

+1 240 501 3163 | utkarsht@umd.edu | utkarsh4430.github.io

LinkedIn | Google Scholar

Current Research & Experience

Scale AI San Francisco, California

Machine Learning Research Engineer *June 2025 – Present*

- Research **audio and multimodal AI**, including speech-to-speech systems and multimodal reasoning, through supervised fine-tuning, reinforcement learning, reward design, and evaluation.
- Developed **SteerDuplex**: SFT and two-stage RL for steerable full-duplex speech dialogue; improved audio-steering average pass rate by **44.5 percentage points** over the strongest evaluated open baseline.
- Developed **POW3R (NeurIPS 2026)**, a policy-aware rubric reward method reaching comparable performance in **2.5–4× fewer training steps**; coauthored **Rubric Dropout** and **Rubric-Guided Self-Distillation**.
- Coauthored **Audio MultiChallenge (ACL Best Resource Paper)**, **Humanity's Sixth Sense**, **Beyond Seeing**, and **SciPredict**, evaluating spoken dialogue, intuitive visual reasoning, visual tool use, and scientific prediction.
- **RSI Bench verified reviewer**: select multimodal and infrastructure tasks; strengthen evaluation protocols for video inference, speculative decoding, KV-cache pruning, and LLM routing.
- Coauthored **RSI-Forge**, a pipeline turning papers into **210 executable research environments across 18 fields**, including 90 independently reviewed by domain experts.
- Define technical specifications, difficulty analysis, and quality standards for multimodal data programs; build annotation tooling and containerized evaluations for GUI grounding, agentic software engineering, and visual reasoning.

Selected Publications

Full publication list: Google Scholar. Paper titles link to the manuscripts.

- SteerDuplex: Steerable Duplex Speech Dialogue Models
  NeurIPS 2026 Workshop RTCA | Under submission ARR
- Not Every Rubric Teaches Equally: Policy-Aware Rubric Rewards for RLVR
  NeurIPS 2026
- RSI-Forge: From Research Papers to Environments for Recursive Self-Improvement
  ICLR 2027 (under submission)
- Humanity's Sixth Sense: Benchmarking Intuitive Visual Reasoning in Multimodal Models
  ICLR 2027 (under submission)
- Rubric Dropout: A Simple Way to Mitigate Reward Hacking in Rubric-Based RL
  ICLR 2027 (under submission)
- Rubric-Guided Self-Distillation: Post-Training Without Rubric Verifiers
  ICLR 2027 (under submission)
- Audio MultiChallenge: A Multi-Turn Evaluation of Spoken Dialogue Systems on Natural Human Interaction
  ACL 2026 | Best Resource Paper Award
- Beyond Seeing: Evaluating Multimodal LLMs on Tool-Enabled Image Perception, Transformation, and Reasoning
  EMNLP 2026
- SciPredict: Can LLMs Predict the Outcomes of Scientific Experiments in Natural Sciences?
  ICML 2026
- Audio Hallucination Attacks: Probing the Reliability of Large Audio Language Models
  Interspeech 2026
- MMAU: A Massive Multi-Task Audio Understanding and Reasoning Benchmark
  ICLR 2025 | Spotlight
- VDGD: Mitigating LVLM Hallucinations in Cognitive Prompts by Bridging the Visual Perception Gap
  ICLR 2025
- ProSE: Diffusion Priors for Speech Enhancement
  NAACL 2025
- GAMA: A Large Audio-Language Model with Advanced Audio Understanding and Complex Reasoning Abilities
  EMNLP 2024
- CompA: Addressing the Gap in Compositional Reasoning in Audio-Language Models
  ICLR 2024

Additional Industry Experience

Samsung SARC/ACL — San Jose, California
Research Intern – Efficient GenAI Inference — *May 2024 – August 2024*

- Optimized LLMs for resource-constrained devices, improving inference latency and memory footprint; fine-tuned models and developed retrieval-augmented generation tools over internal documentation.

Atlassian India LLP — Bangalore, India
Software Development Engineer 2 — *July 2021 – August 2023*

- Modernized Jira's issue-create experience, achieving **99.996% frontend reliability** for over **250,000 customers** and **2 million issues daily**; reduced time to interactive from **3.9s to 1.2s**.

Atlassian India LLP — Bangalore, India
Software Engineer Intern — *January 2021 – June 2021*

- Reworked legacy APIs and eliminated redundant database queries, reducing modal-load latency by **95.57%** and supporting a user interface serving **1 million customers**.

Samsung R&D Institute Bangalore — Bangalore, India
Research Intern – Speech AI — *May 2020 – July 2020*

- Researched Bixby wake-word detection and developed an efficient, vocabulary-independent on-device keyword spotter using **LSTM-CTC**.

Academic Research Experience

GAMMA Lab, University of Maryland — College Park, Maryland
Research Assistant — *August 2022 – May 2025*

- Researched audio-language models, multimodal reasoning, speech enhancement, and low-resource learning with **Prof. Dinesh Manocha** and **Prof. Ramani Duraiswami**; work includes **MMAU**, **GAMA**, **CompA**, and **ProSE**, plus synthetic data augmentation for speech and NLP.

Multimodal Digital Media Analysis Lab, IIITD — Delhi, India
Machine Learning Researcher — *April 2022 – March 2023*

- Worked with **Dr. Rajiv Ratn Shah** on multilingual automatic speech scoring for low-resource Indian languages and context-aware implicit hate-speech detection; published at the **De-Factify workshop, AAAI 2023**.

Education

University of Maryland, College Park — College Park, Maryland
M.S. in Computer Science | GPA: 3.9/4.0 — *August 2023 – May 2025*

Delhi Technological University — Delhi, India
B.Tech in Computer Science | GPA: 8.7/10; major GPA: 9.05/10 — *2017 – 2021*

Technical Skills

Research: Speech-to-speech; full-duplex dialogue; audio-language models; multimodal reasoning; benchmark design.

Post-training: SFT; reinforcement learning (GRPO/RLVR); rubric rewards; self-distillation; reward-hacking analysis.

Languages & Frameworks: Python, C++, JavaScript, Java, SQL; PyTorch, TensorFlow, scikit-learn, React.

Research Engineering: Evaluation harnesses; annotation tools; containerized environments; rubric grading; data quality.

Awards & Professional Service

- **ACL 2026 Best Resource Paper Award** – Audio MultiChallenge (shared with coauthors).
- **$7,000 NeuroPAC Fellowship**, UMD/UMIACS – research on EEG-based brain-computer interaction using LLMs.
- **Peer reviewer:** NeurIPS 2026; ICLR 2025–2026; ACL Rolling Review 2024–2026 (including recent cycles); ICASSP 2025 and 2027 (current cycle); WASPAA 2025; EMNLP 2024.
- **Atlassian internal hackathon People's Choice Awards**, 2021 and 2022.

Patents

- **U.S. Patent 12,505,292 B2** – scheduling abstractive summaries for multi-party communication channels; co-inventor, granted December 23, 2025.
- **U.S. Application 17/936,705 (US20240111959A1)** – generating and selectively outputting abstractive summaries; co-inventor, patent pending.
- **U.S. Application 18/477,176 (US20250110981A1)** – configuring and transmitting abstractive summaries to collaboration applications; co-inventor, patent pending.